EE/CprE/SE 491 WEEKLY REPORT V

Group number: 40

Project title: Digital ASIC Fabrication

Client &/Advisor: Dr. Duwe

Team Members/Role: David Rice | *Project Management lead*, Ian Runestad | *Software lead*, Dhruv Patel | *Communication lead*, Jay Patel | *FPGA verification lead*; Christopher Funk | *Documentation lead*

Date: 10/6/2026 – 10/13/2026

- **Weekly Summary**
    - The focus of this week's work involved working on the PID controller (David), evaluating and choosing communication modules (Christopher) and RISC-V processor IP cores (Dhruv) based on our application's needs, and setting concrete deadlines for when the various parts of the design should be completed.
- **Past week's accomplishments**
    - David Rice - SV unit testbenches pass with randomized testing for nested cases of PID units. Am able to start up RTL simulations for design put inside of user project wrapper which is needed for FPGA testing and synthesis for area estimation.
    - Ian Runestad - Looked into viable IP blocks for UART functionality, prioritizing compatibility and verifiable tapeout history. No chosen core as of 10/13.
    - Dhruv Patel - Made changes to Processor and the interfaces for integrating the Ibex with the Chipforge toolchain. Also looked at the required processor interface,

memory connections, and Wishbone bus integration to prepare the Ibex processor for simulation and testing.

- Jay Patel - Looked into the four-channel PWM design and how to use one counter for different duty cycles. Also reviewed the Ibex processor interfaces to understand what needs to be tested on the FPGA.
- Christopher Funk - learned how to configure and flash the Arty A7-100T FPGA both with the Caravel user project and a custom Verilog module. Chose IP cores for the microcontrollers' SPI and I2C interfaces and imported these IP cores into my Git branch. Began learning about how the Wishbone and UL-TL buses work and started to design a TL-UL to Wishbone bridge so that the SPI and I2C IP cores can interface with the other devices in the microcontroller.

- **Pending issues**
  - David Rice - Expanding the functionality of RTL testing so it can be deployed on FPGA and hardened for area estimation, though some of the tools have been updated and not very thoroughly tested.
  - Ian Runestad - N/A
  - Dhruv Patel - Need to verify the processor compatibility with the Chipforge Toolchain.
  - Jay Patel - Need to finalize the PWM frequency and wait for the Ibex integration before starting FPGA testing.
  - Christopher Funk - N/A
- **Individual contributions**

NAME	Individual Contributions	Hours this week	HOURS cumulative

David Rice	Work on nested PID units and getting reading for FPGA development and hardening	7	37
Ian Runestad		2	11
Dhruv Patel	Worked on Ibex processor configuration and Chipforge integration.	4	24
Jay Patel	Worked on PWM design and looked into FPGA verification requirements for the Ibex processor.	4	20
Christopher Funk	Worked on integrating chosen I2C and SPI controllers with the Wishbone bus. Practiced configuring and flashing the FPGA.	6	27

- **Plans for the upcoming week**
    - David Rice - Get FPGA synthesis to start, and work on area estimation script when I'm around Chipforge, since the previous year's members have some experience with that.
    - Ian Runestad - Choose UART IP core, validate it on an actual design, test loopback between two UART IPs.
    - Dhruv Patel - Finish setting up the Ibex processor and run basic simulation tests. Based on next Thursday's discussion with Dr. Duwe, either continue with Ibex or set up a different processor.

    - Jay Patel - Plan to start writing PWM testbenches and work on FPGA testing for the Ibex processor.
    - Christopher Funk - Finish writing the TL-UL to Wishbone bridge, begin writing testbenches verifying that the TL-UL to Wishbone bridge works correctly.

- **Summary of weekly advisor meeting**

In our meeting with Dr. Duwe, the following points were discussed the following points:

- Drive the clock frequency from application requirements, since meeting the final timing constraints matters more than any specific frequency.
- Consider a single clock domain for verification and design-complexity reasons, recognizing that the processor may then limit how aggressive the PID controller clock can be.
- Do not expect the processor itself to need a particularly high clock frequency.
- Determine the minimum PWM frequency needed for stable motor operation by researching the specific motors, and favor measuring a physical device over relying on datasheet values.
- Try out claims found in documentation quickly before trusting them, as some features may be much harder to implement than described.
- Select OpenTitan as the leading I2C/SPI IP candidate because it is the most fully featured, and consider adding clock gating later since it is not overly complex.
- Plan to create Wishbone wrappers for most candidate IPs, and note that ChipFoundry IP has an advantage because it has been fabricated in the 130 nm process and is likely the most reliable.
- Add Bosch IMU compatibility to the requirements, including the SPI 16-bit data frame and the lack of I2C clock-stretching support, as a justification for IP choice.
- Note that OpenTitan and PULP (ETH Zurich) support only multi-byte SPI frames, not arbitrary bit-length frames, and evaluate whether this matters.
- Expand the SPI comparison with more rows covering configuration options such as clock polarity/phase and data sampling timing, since SPI is not a formal standard and IP support is unclear.

- Test candidate IP cores on FPGA, ideally with the actual target devices such as the Bosch IMU (via header pins or a breakout board, possibly borrowed from the course robot hardware).
- Present a feature-comparison table across IP cores to show the choice reflects real consideration of features, not just the first IP found, and include documentation quality (strongest for OpenTitan) and silicon-proven status.
- Evaluate whether 32-bit precision is needed, since 16-bit or 24-bit may be sufficient depending on sensor precision and the number of computations.
- Account for every pin on the Caravel user harness (each must be driven) so the design passes pre-check, which also supports FPGA testing.
- Update the area-estimation script to work with the newer tools.
- Reuse the existing Efabless Caravel UART instead of building one, and check whether Efabless IP is still supported by ChipFoundry.
- Treat reuse of proven IP as fully acceptable and favorably viewed when justified with tables and engineering analysis, since the PID design, software, verification, and GDSII flow still provide sufficient complexity.
- Bring a draft of slides, even if unfinished, to every meeting, and meet regularly outside the team meeting to coordinate and give feedback on slides.
- Reschedule the October 22 meeting, as an 8:30 AM start was not workable for the team.